

Understanding Data Scarcity in Foundation-Model-Based Histopathology Patch Classification

Yannik Queisler

Thesis Proposal

October 9, 2026

1 Motivation

Pathology foundation models provide transferable representations for histopathology patch classification (Chen et al., 2024; Vorontsov et al., 2024). A classifier can then learn a diagnostic task from these representations without retraining the large model. Public benchmarks compare the models on clinical tasks (Campanella et al., 2025), but offer less guidance on a practical question: **which labeled data does a downstream classifier actually need?** This matters because expert annotation is expensive (Lu et al., 2021), especially for rare cancers.

The thesis distinguishes two forms of scarcity. *Class imbalance* means that common diagnoses receive much more training data than rare ones; this can harm both recognition and the reliability of predicted probabilities (Mosquera et al., 2024). *Data shortage* means having too little labeled evidence overall, either too few patches or too few independent patients. These are different limits: more patches from one patient may add tissue detail without representing the variation between patients. Patient separation is also essential for credible evaluation, since mixing a patient’s patches across training and testing can inflate performance (Bussola et al., 2021).

The study uses foundation models as backbones and examines both data scarcity and whether the choice of backbone changes its effects. The aim is to explain when annotation effort should target rare classes, more tissue per patient, or additional patients, and when an algorithm can compensate.

2 Research Questions

Both class imbalance and data shortage are examined through three questions: how much performance they cost, what causes the loss, and how effectively it can be mitigated. This progression connects the measured effects to explanations and practical remedies (Figure 1).

1. **Performance impact:** How do class imbalance and data shortage affect discrimination and probability quality across tasks, and how do patch count and patient count differ in their effects?
2. **Mechanisms:** What causes these performance losses, why are some classes and tasks more vulnerable, and why can additional patches fail to replace additional patients?
3. **Mitigation:** How effectively do mitigation methods improve performance under class imbalance and data shortage, and why do some methods work better than others?

3 Methodology

Controlled experiments use 30 cancer types from TCGA-UT (Komura et al., 2022) and seven breast-lesion classes from BRACS (Brancati et al., 2022). A linear classifier on frozen features provides the baseline; matched experiments compare Virchow2 (Zimmermann et al., 2024) and UNI2-h to test the relevance of backbone choice. Patients stay separate across training, validation and testing.

Figure 1 summarizes the design. For **class imbalance**, class shares change at fixed patch budget and patients per class. For **data shortage**, patch count changes at fixed patient coverage, then patient count at a matched patch budget, with class allocation controlled. Recognition gives equal weight to classes and patients; calibration checks confidence against accuracy. Paired comparisons across patient splits quantify uncertainty.

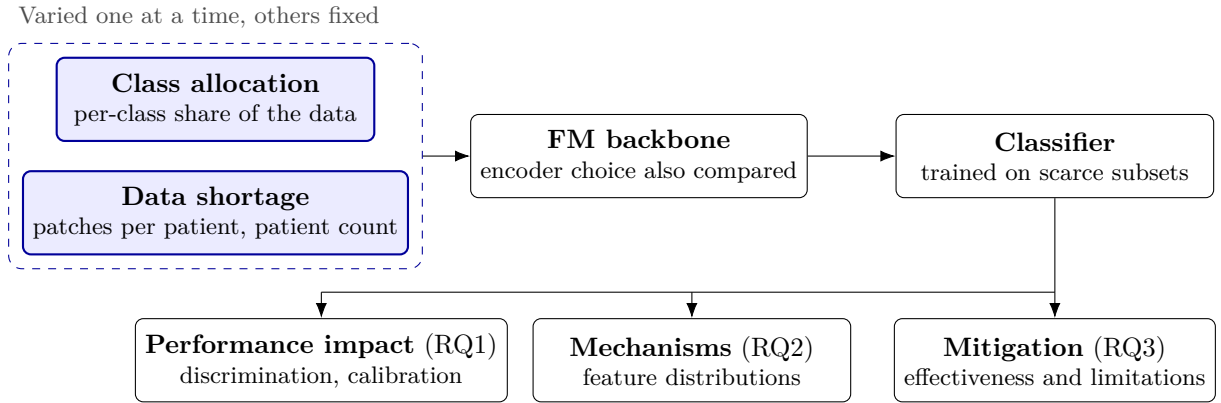


Figure 1: Experimental design. Each scarcity factor (blue) is varied alone while the others stay fixed.

4 Preliminary Findings

Initial class-imbalance experiments examine performance impact and underlying mechanisms on both tasks, using Virchow2 and a matched comparison with UNI2-h. These experiments vary the largest-to-smallest class ratio ρ from 1 to 100 over 30 combinations of patient split and class allocation (Figure 2). Three findings guide the remaining work.

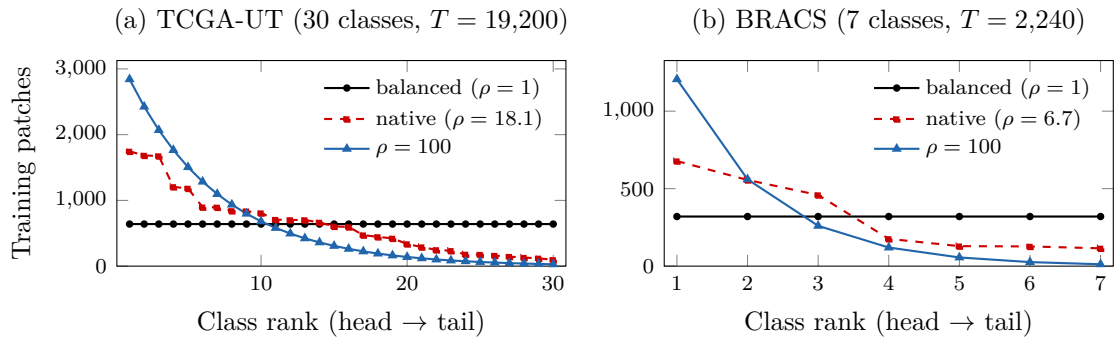


Figure 2: Training class distributions: balanced, native and ratio 100, sorted from most to least frequent. Total patch budget T and patients per class stay fixed; only class shares change.

1. Which class is rare matters as much as the ratio. On TCGA-UT at ratio 100, changing which classes receive few patches changes balanced-accuracy loss from 1.3 to 4.5 percentage points (pp). Damage is greatest when a rare class is often mistaken for a class that receives many examples: predictions favor the common alternative. On BRACS, classes that are easy to recognize with balanced data have the most to lose when made rare. The ratio alone cannot identify which diagnoses need more labels.

2. Probability quality is easier to improve than discrimination. At ratio 100, expected calibration error (ECE), the gap between confidence and accuracy, rises from 5.5 to 17.8 pp on TCGA-UT and from 12.3 to 35.7 pp on BRACS (Figure 3b). Temperature scaling, one validation-fitted confidence adjustment (Guo et al., 2017), removes 92–96% of the imbalance-induced increase relative to the corrected balanced baseline. Predicted classes stay unchanged: better confidence does not recover missed diagnoses.

3. Imbalance damage differs strongly between tasks. TCGA-UT tolerates moderate imbalance: balanced accuracy falls by only 0.6 pp at ratio 10 and 2.9 pp at 100. BRACS is more sensitive, losing 1.2 pp already at ratio 2 and 7.7 pp at 100 (Figure 3a). At ratio 100, its two rarest classes fall from 38.7% to 7.3% recall: most of their patches are misclassified. The tasks differ in what must be distinguished: TCGA-UT spans cancers from different organs, whereas BRACS separates breast lesions with similar morphology. This makes the greater sensitivity of BRACS plausible. Experiments also show two sources of damage: rare classes have less influence during training and fewer examples of their variation. Giving those examples more weight addresses the first problem, but cannot replace the missing examples. The pattern persists with UNI2-h, so changing backbone alone does not solve it.

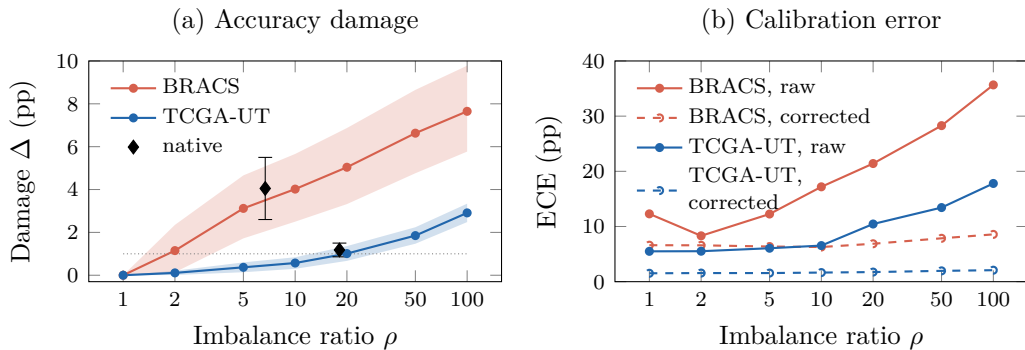


Figure 3: (a) Balanced-accuracy loss (percentage points, pp), with 95% intervals; diamonds show native class shares and the dotted line marks 1 pp. (b) Calibration error before and after temperature scaling: probability quality is easier to improve than discrimination.

5 Next Steps

The thesis will extend the imbalance findings to **data shortage**, then evaluate **mitigation methods for both forms of scarcity**. The aim is to establish which methods work well, where they fall short, and why, alongside guidance on where additional labels are most valuable.

The remaining work follows two stages:

- **Data shortage.** Reduce patches at fixed patient coverage, then patients at a matched patch budget. Measure the performance impact and explain whether tissue variation or patient diversity limits performance, and when more patches stop helping.
- **Mitigation.** Evaluate methods for class imbalance and data shortage, including class reweighting, estimation from few examples, and temperature scaling. Compare their effectiveness across tasks and scarcity levels, and analyze why some methods succeed while others offer little improvement.

References

- Nadia Brancati, Anna Maria Anniciello, Pushpak Pati, Daniel Riccio, Giosue Scognamiglio, Guillaume Jaume, Giuseppe De Pietro, Maurizio Di Bonito, Antonio Foncubierta, Gerardo Botti, Maria Gabrani, Florinda Feroce, and Maria Frucci. BRACS: A dataset for BReAst carcinoma subtyping in H&E histology images. *Database*, 2022:baac093, 2022. doi: 10.1093/database/baac093.
- Nicole Bussola, Alessia Marcolini, Valerio Maggio, Giuseppe Jurman, and Cesare Furlanello. AI slipping on tiles: Data leakage in digital pathology. In *Pattern Recognition. ICPR International Workshops and Challenges*, 2021. URL <https://arxiv.org/abs/1909.06539>.
- Gabriele Campanella et al. A clinical benchmark of public self-supervised pathology foundation models. *Nature Communications*, 16, 2025. doi: 10.1038/s41467-025-58796-1.
- Richard J. Chen, Tong Ding, Ming Y. Lu, et al. Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3):850–862, March 2024.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR, 2017.
- Daisuke Komura, Akihiro Kawabe, Kenta Fukuta, Kazuya Sano, Tsukasa Umezaki, Haruka Koda, Riku Suzuki, Keisuke Tominaga, Michiaki Ochi, Hirofumi Konishi, et al. Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424, 2022. doi: 10.1016/j.celrep.2022.110424.
- Ming Y. Lu et al. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering*, 5(6):555–570, 2021. doi: 10.1038/s41551-020-00682-w.
- Candelaria Mosquera, Luciana Ferrer, Diego H. Milone, Daniel Luna, and Enzo Ferrante. Class imbalance on medical image classification: towards better evaluation practices for discrimination and calibration performance. *European Radiology*, 34(12):7895–7903, 2024. doi: 10.1007/s00330-024-10834-0.
- Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, Eric Zimmermann, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935, 2024. doi: 10.1038/s41591-024-03141-0.
- Eric Zimmermann, Eugene Vorontsov, Julian Viret, et al. Virchow2: Scaling Self-Supervised Mixed Magnification Models in Pathology. *arXiv preprint arXiv:2408.13910*, 2024.